



UNIVERSITY OF SINDH
JAMSHORO

MINI ASSIGNMENT

DECONSTRUCTING HCI & CG

IN DAILY SOFTWARE

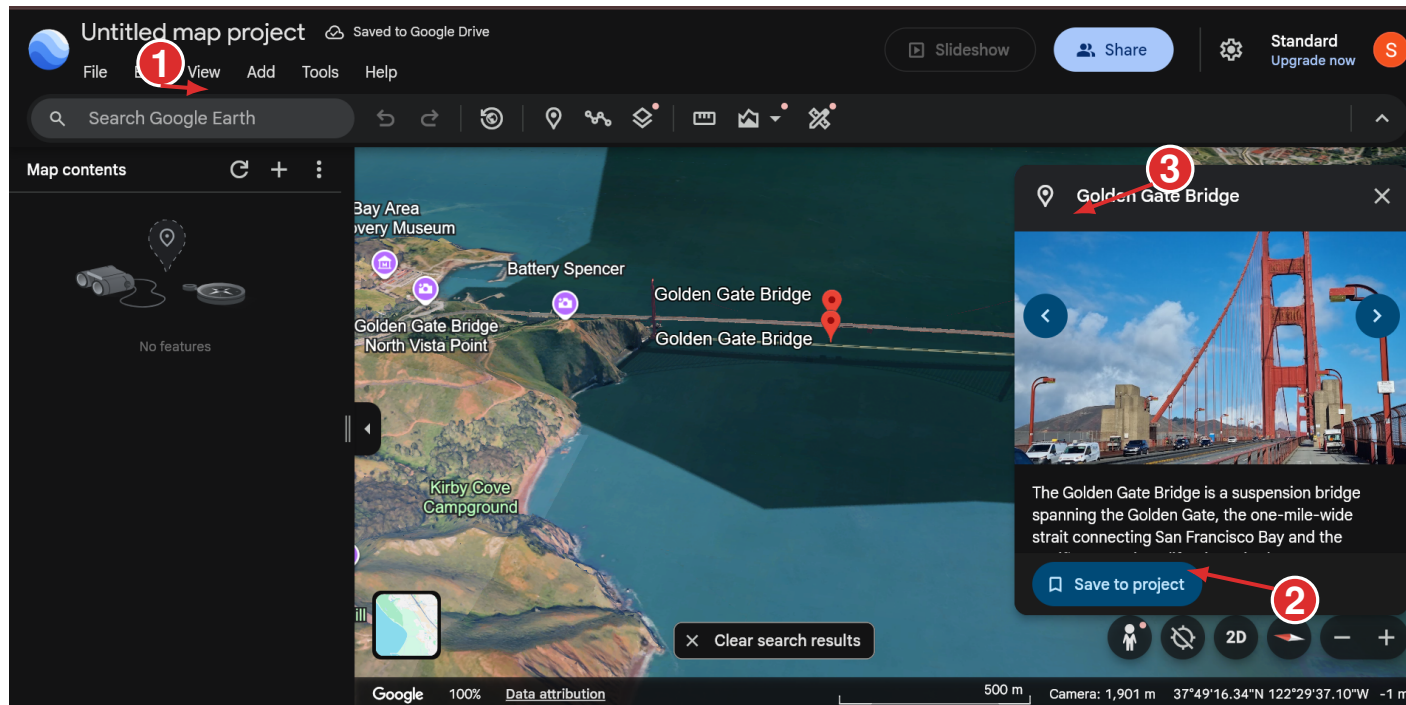


DEPARTMENT OF
COMPUTER SCIENCE

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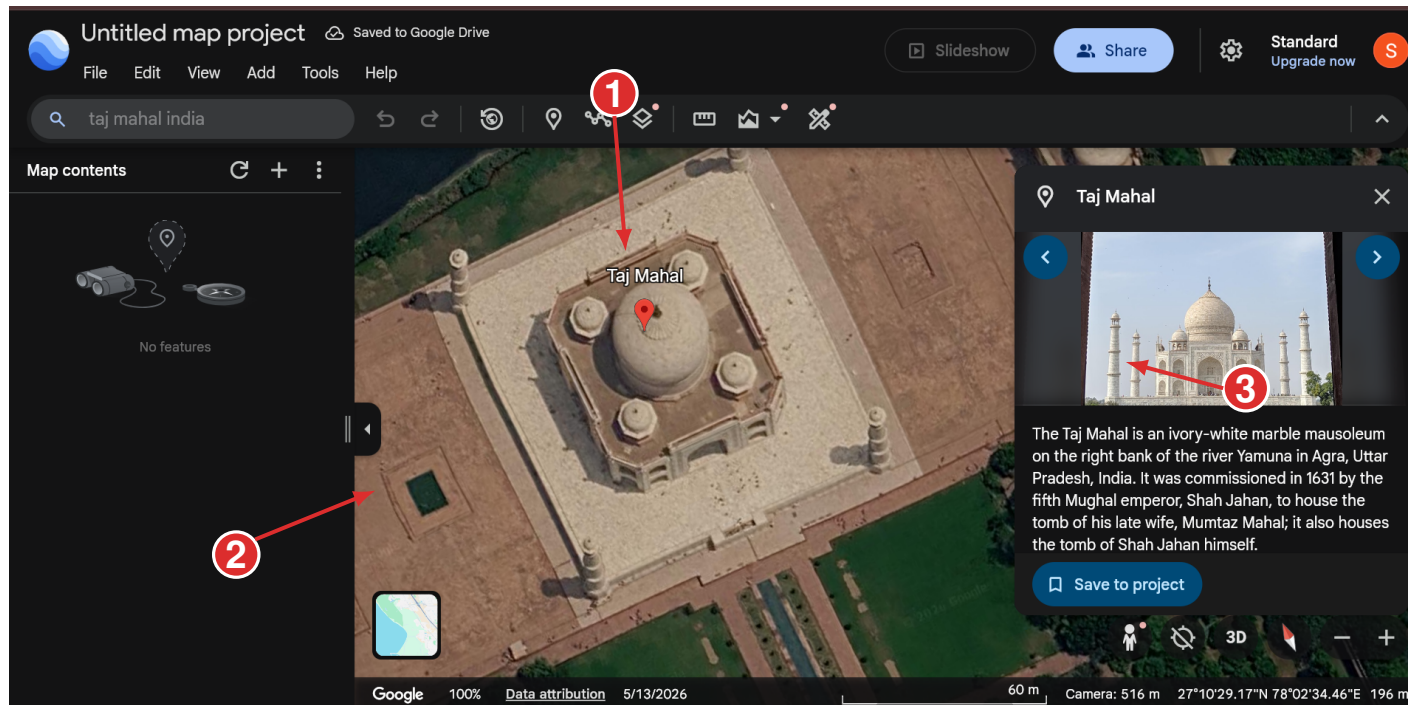


SCREENSHOT 1: HCI FOCUS



- 1. Search Bar (Top-Left):** Allows users to quickly search for any location worldwide. The search function provides a direct way to enter a destination and locate it on the map, reducing the effort required to find places.
- 2. 3D Navigation Controls (Bottom-Right):** Includes zoom, rotation, compass, and view controls that let users explore the 3D environment from different angles. These controls provide clear visual affordances for manipulating the map view.
- 3. Information Panel (Right-Side):** Displays information about the selected location, including its name, image, description, and the option to save the place to the project. This provides immediate contextual feedback after selecting a landmark.

SCREENSHOT 2: CG FOCUS



- 1. 3D Mesh Geometry:** The Taj Mahal is represented through detailed three-dimensional architectural geometry. The central dome, surrounding structures, and minarets create recognizable 3D forms and help represent the landmark in three-dimensional space.
- 2. Texture Mapping:** Detailed satellite imagery provides realistic visual textures for the Taj Mahal, gardens, pathways, water, and surrounding environment. These surface details improve the visual realism and context of the 3D scene.
- 3. Perspective & Depth:** The elevated camera position, perspective, object scale, and spatial arrangement create depth cues. These visual elements help users understand the relative position, size, and distance of objects in the 3D environment.

REFLECTION QUESTIONS

Question 1: How does the Computer Graphics quality (e.g., realistic lighting or 3D view) help or hinder the user's ability to interact with the software?

The high-quality computer graphics in Google Earth, including detailed 3D geometry, realistic textures, perspective, and depth cues, significantly enhance the user's ability to navigate and understand spatial relationships. The visual realism makes it easier to recognize landmarks and orient oneself within the 3D space, which directly supports effective interaction. However, rendering complex graphics requires substantial processing power, which can cause performance issues on older or less powerful devices and may hinder smooth interaction.

Question 2: What is one simple HCI change you would make to improve the user interface?

One simple HCI improvement would be to add a customizable "Quick Actions" toolbar that allows users to pin their most frequently used tools. Important functions such as measuring distance, switching to Street View, and changing 3D viewing options can require users to navigate through menus. A pinned toolbar would make frequently used tools accessible with a single click, reducing interaction cost and improving overall efficiency.

Grading Breakdown

Component	Requirement	Points
HCI Annotation	Correctly identifies 3 interaction/UI elements and their usability purpose.	35
CG Annotation	Correctly identifies 3 graphics/rendering elements such as geometry, texture, perspective, lighting, or depth.	35
Reflection	Clear and concise answers connecting computer graphics and rendering to the user experience.	20
Clarity & Layout	Clean screenshot layout with easily readable callouts and pointers.	10
TOTAL		100